

CS 4033/5033 Project 3

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1 Clustering methods

We chose K-Means clustering because it is simple and works well for grouping similar data. It splits the data into a set number of clusters based on how close points are to a center. It is also quick to run and easy to understand, which made it a good fit for our project.

2 Goodness measure

As goodness measure we chose Silhouette Score which ranges from -1 to 1. Higher values mean better-defined clusters. We tested this score for k clusters between 2 and 8 and chose the k with the best score.

3 Graphical representation

We plotted the data using 2d and 3d scatter plots while the color of the points represent the selected cluster. Additionally, we plotted the Silhouette Score of each dataset for k and for $k - 1$ and $k + 1$.

4 Clustering

4.1 Dataset 1

Dataset 1 is best represented with 2 clusters. Figure 1 shows that the silhouette score for $k = 2$ is highest and everything greater than $k = 3$ has a bad score. The different clusters for $k = 2$ can be seen in Figure 2 and for $k + 1 = 3$ in Figure 3.

4.2 Dataset 2

Dataset 2 is best represented with 2 clusters. Figure 4 shows that the silhouette score for $k = 2$ is highest and everything greater than $k = 3$ has a bad score. The different clusters for $k = 2$ can be seen in Figure 5 and for $k + 1 = 3$ in Figure 6.

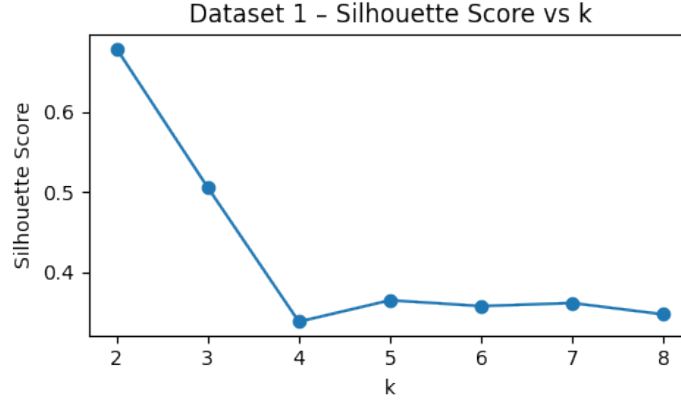


Figure 1: Dataset 1 Silhouette Score vs k

4.3 Dataset 3

Dataset 3 is best represented with 8 clusters. Figure 7 shows that the silhouette score for $k = 2$ is relatively high but the optimal is reached with higher k values. The difference in the silhouette score between k values of 6, 7 or 8 is minimal. The different clusters for $k - 1 = 7$ can be seen in Figure 8, for $k = 8$ in Figure 9 and for $k + 1 = 9$ in Figure 10.

4.4 Dataset 4

Dataset 4 is best represented with 7 clusters. Figure 11 shows that, similar to Dataset 3, the silhouette score for $k = 2$ is relatively high but the optimal is reached with higher k values. However, compared to Dataset 3 the difference between $k = 2$ and higher k values is only slight. Additionally, the difference in the silhouette score between k values of 6, 7 or 8 is minimal. The different clusters for $k - 1 = 6$ can be seen in Figure 12, for $k = 7$ in Figure 13 and for $k + 1 = 8$ in Figure 14.

4.5 Dataset 5

Dataset 5 is best represented with 4 clusters. Figure 15 shows that the silhouette scores for k between 2 and 6 are relatively high but the optimal is reached at $k = 4$. The different clusters for $k - 1 = 3$ can be seen in Figure 16, for $k = 4$ in Figure 17 and for $k + 1 = 5$ in Figure 18.

4.6 Dataset 6

Dataset 6 is best represented with 2 clusters. Figure 19 shows that the silhouette score for $k = 2$ is highest and everything greater than $k = 3$ has a bad score.

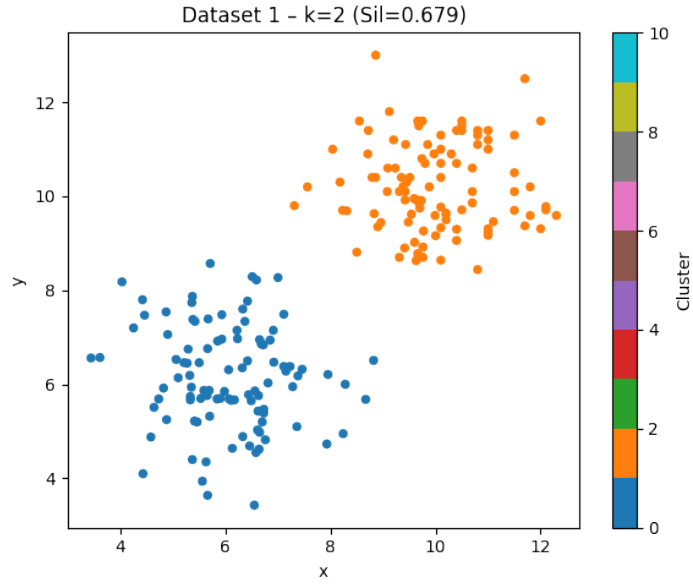


Figure 2: Dataset 1 Clustering with k Clusters

The different clusters for $k = 2$ can be seen in Figure 20 and for $k + 1 = 3$ in Figure 21.

4.7 Dataset 7

Dataset 7 is best represented with 2 clusters. Figure 22 shows that the silhouette score for $k = 2$ is highest and everything greater than $k = 3$ has a bad score. The different clusters for $k = 2$ can be seen in Figure 23 and for $k + 1 = 3$ in Figure 24.

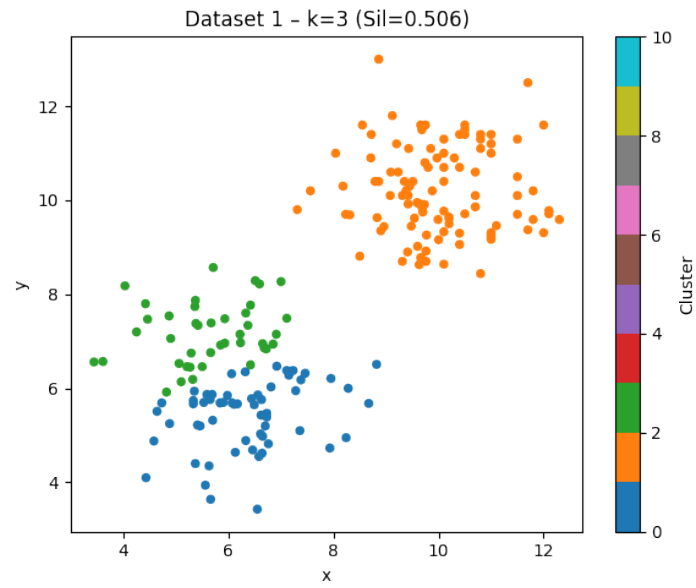


Figure 3: Dataset 1 Clustering with $k + 1$ Clusters

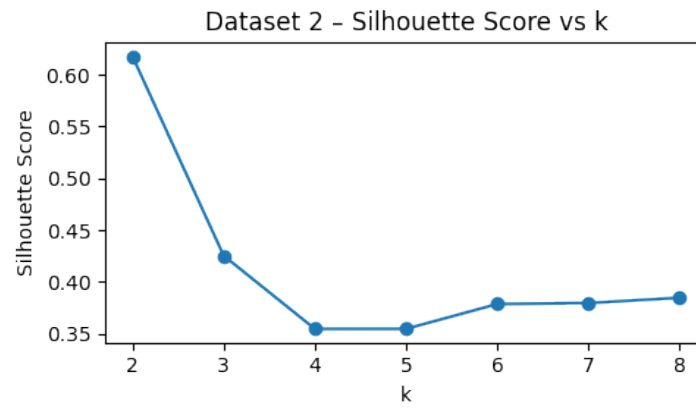


Figure 4: Dataset 2 Silhouette Score vs k

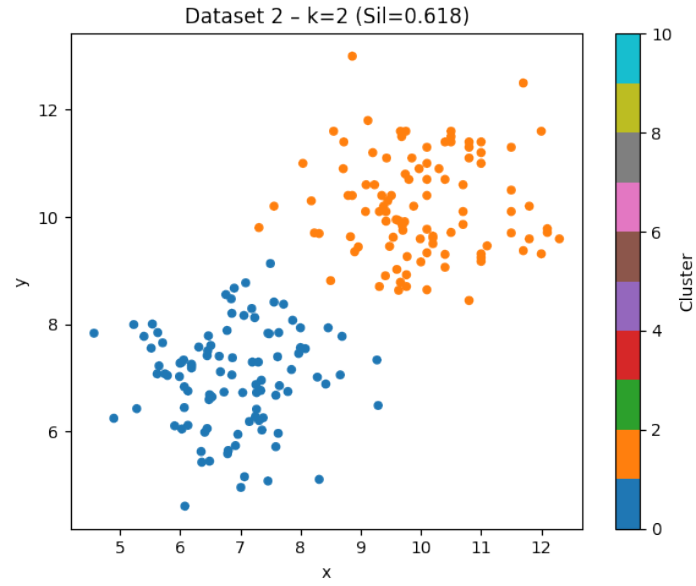


Figure 5: Dataset 2 Clustering with k Clusters

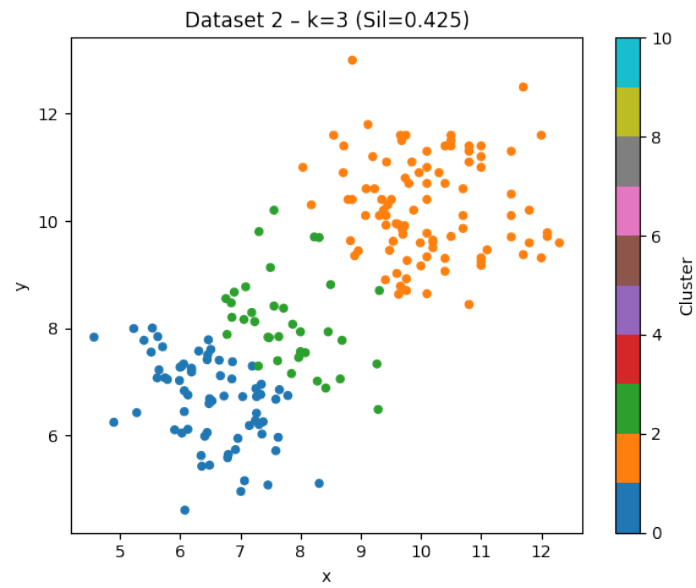


Figure 6: Dataset 2 Clustering with $k + 1$ Clusters

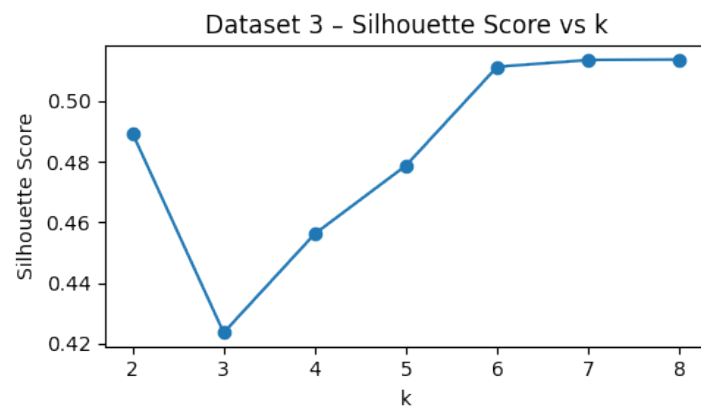


Figure 7: Dataset 3 Silhouette Score vs k

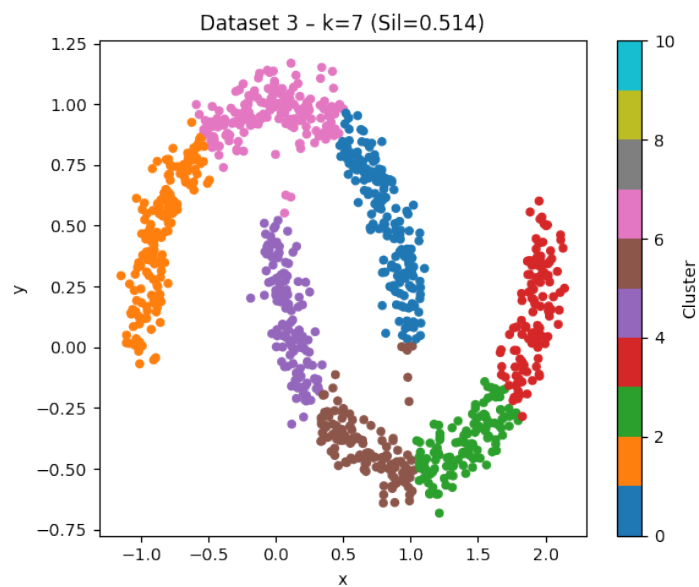


Figure 8: Dataset 3 Clustering with $k - 1$ Clusters

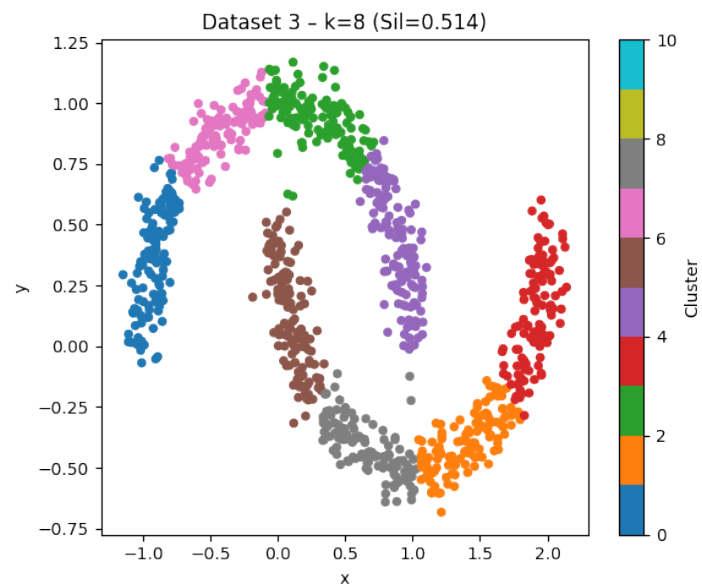


Figure 9: Dataset 3 Clustering with k Clusters

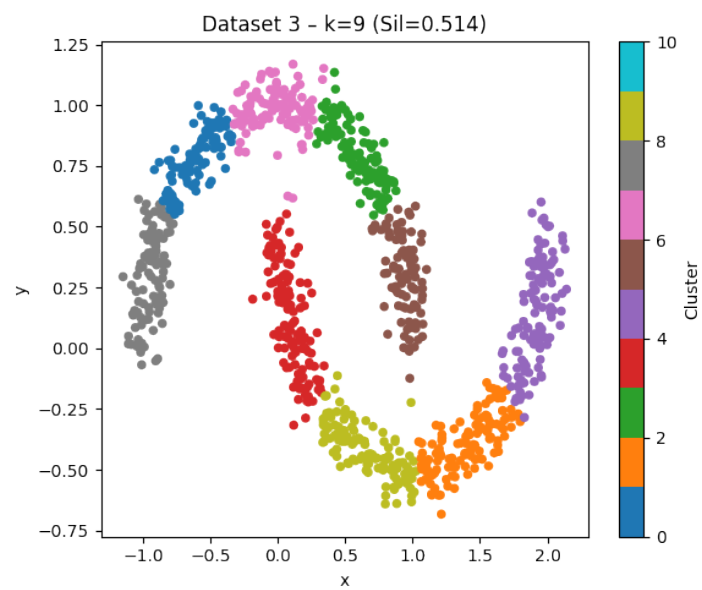


Figure 10: Dataset 3 Clustering with $k + 1$ Clusters

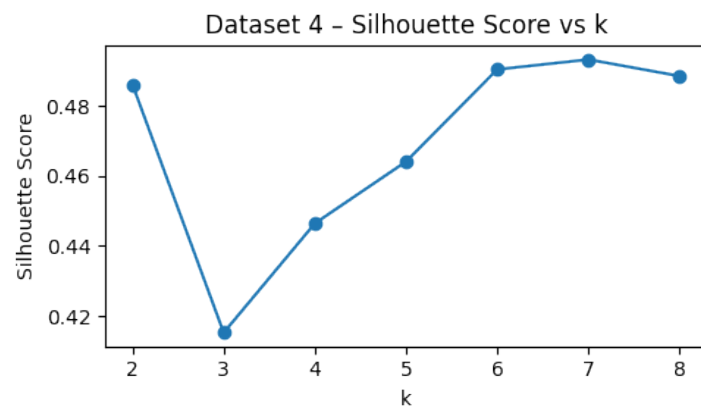


Figure 11: Dataset 4 Silhouette Score vs k

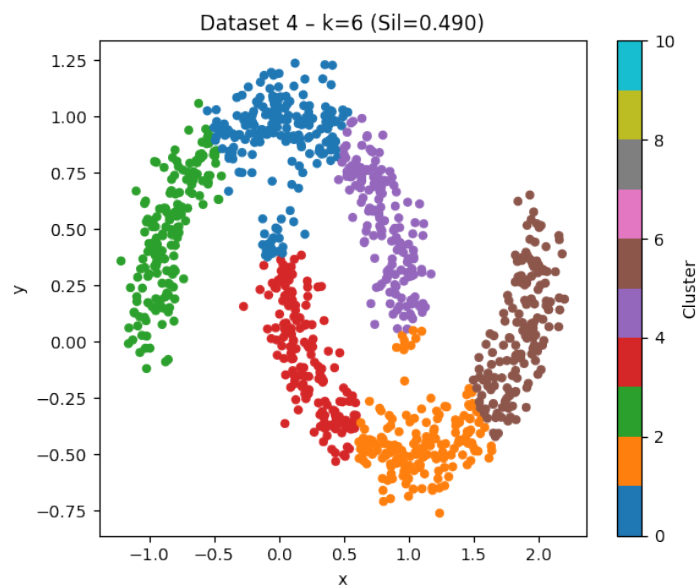


Figure 12: Dataset 4 Clustering with $k - 1$ Clusters

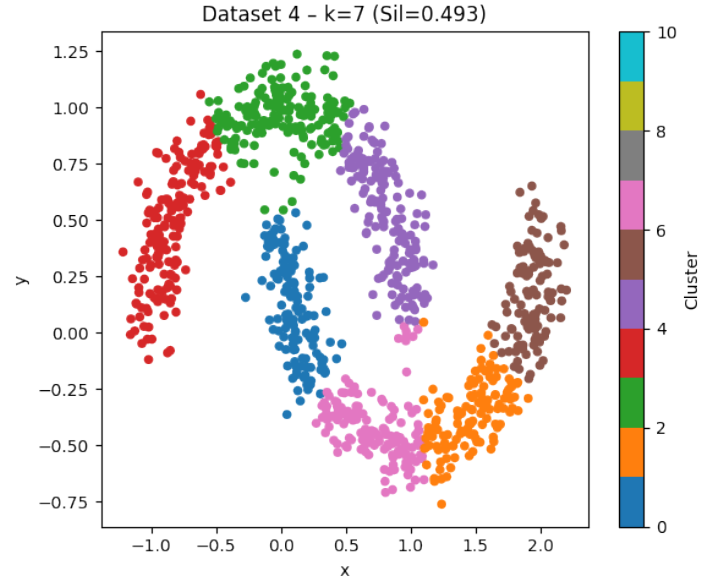


Figure 13: Dataset 4 Clustering with k Clusters

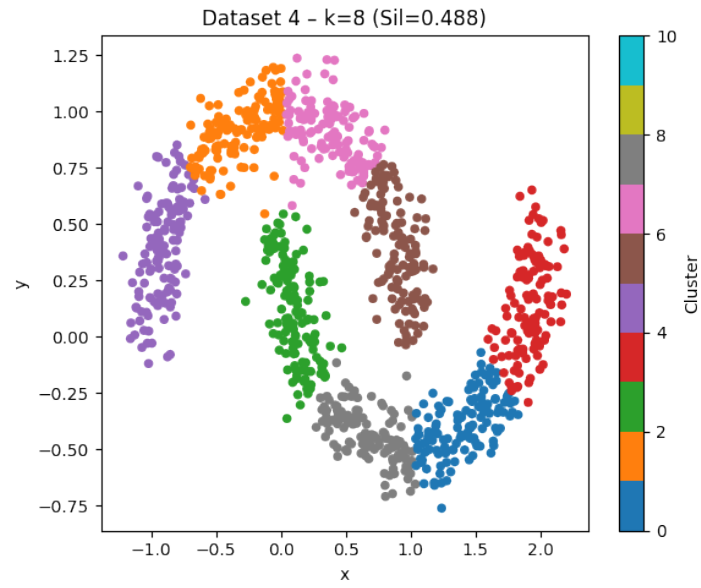


Figure 14: Dataset 4 Clustering with $k + 1$ Clusters

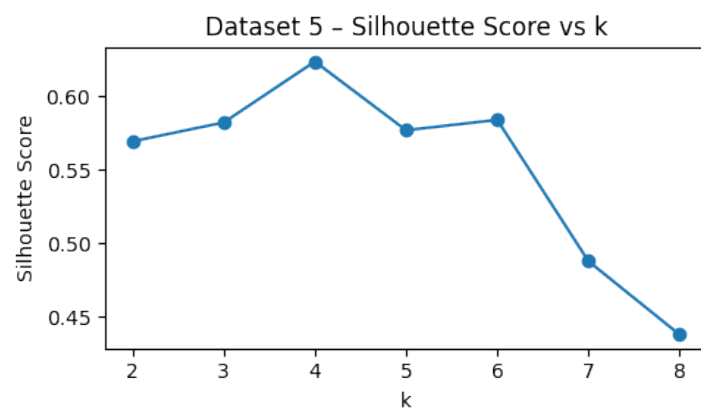


Figure 15: Dataset 5 Silhouette Score vs k

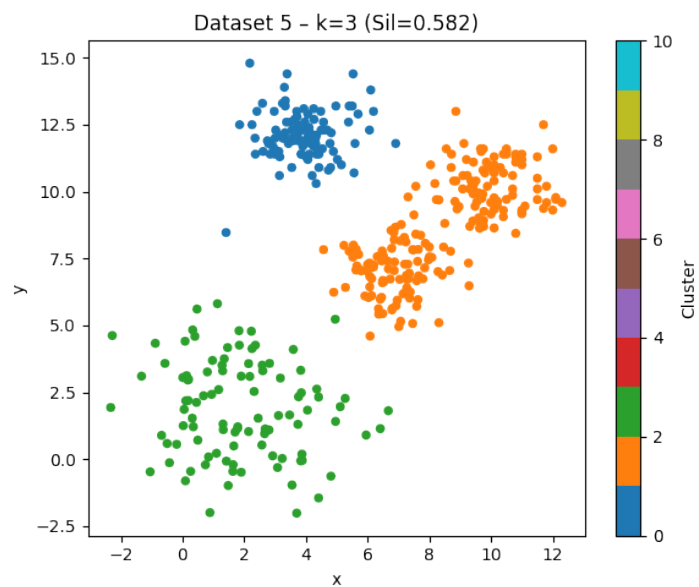


Figure 16: Dataset 5 Clustering with $k - 1$ Clusters

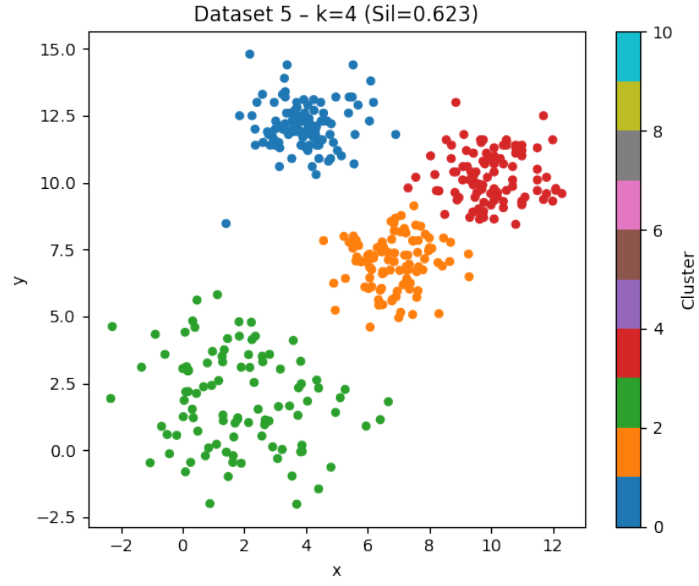


Figure 17: Dataset 5 Clustering with k Clusters

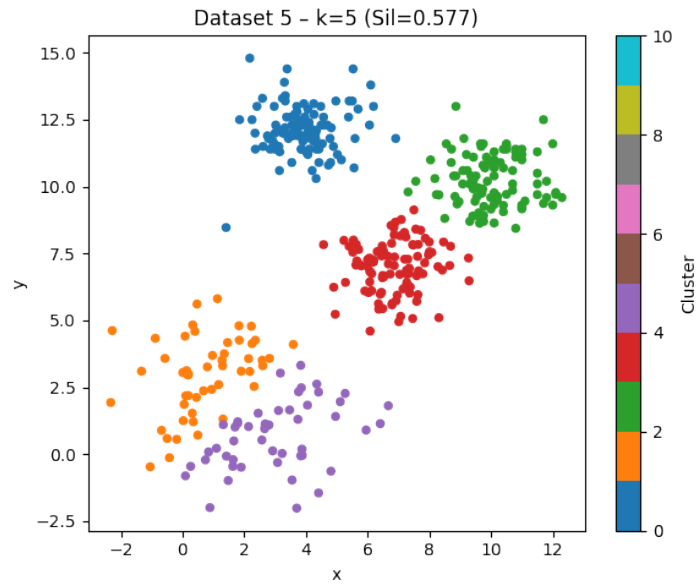


Figure 18: Dataset 5 Clustering with $k + 1$ Clusters

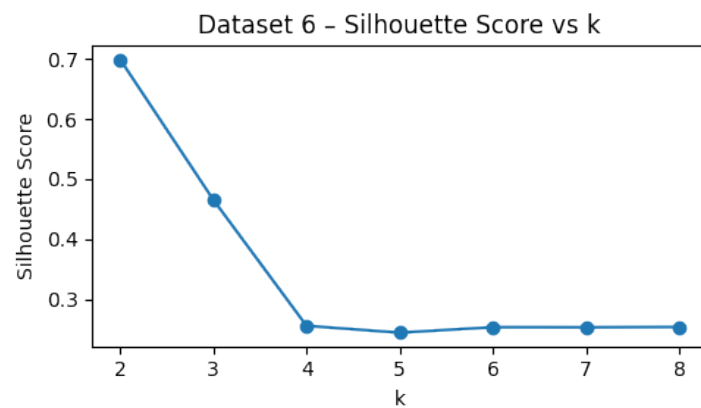


Figure 19: Dataset 6 Silhouette Score vs k

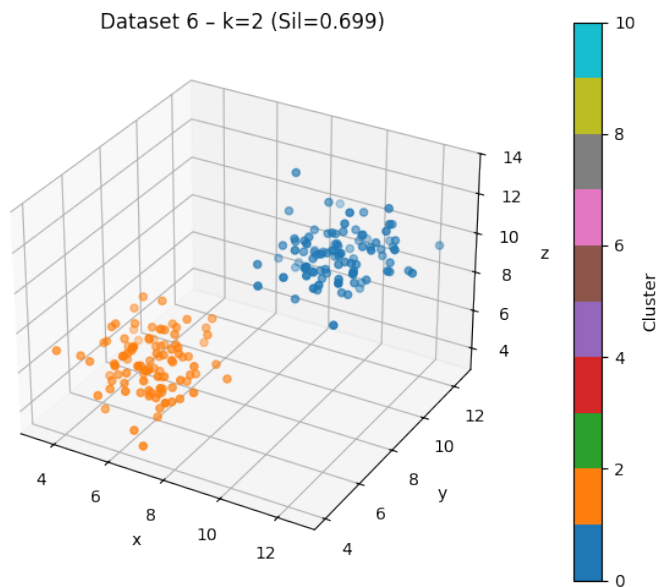


Figure 20: Dataset 6 Clustering with k Clusters

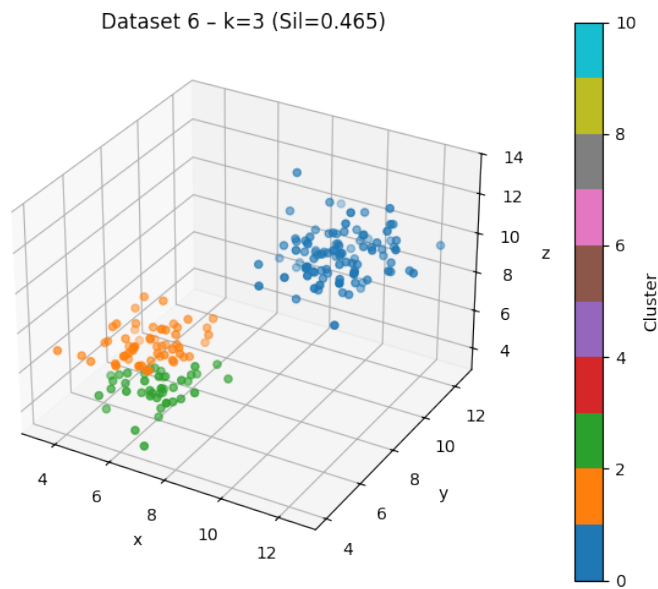


Figure 21: Dataset 6 Clustering with $k + 1$ Clusters

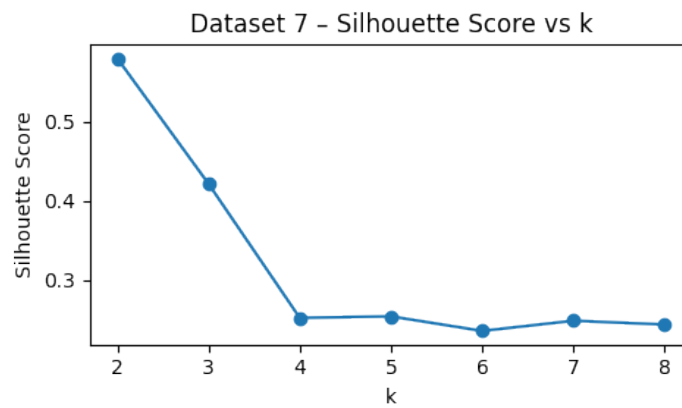


Figure 22: Dataset 7 Silhouette Score vs k

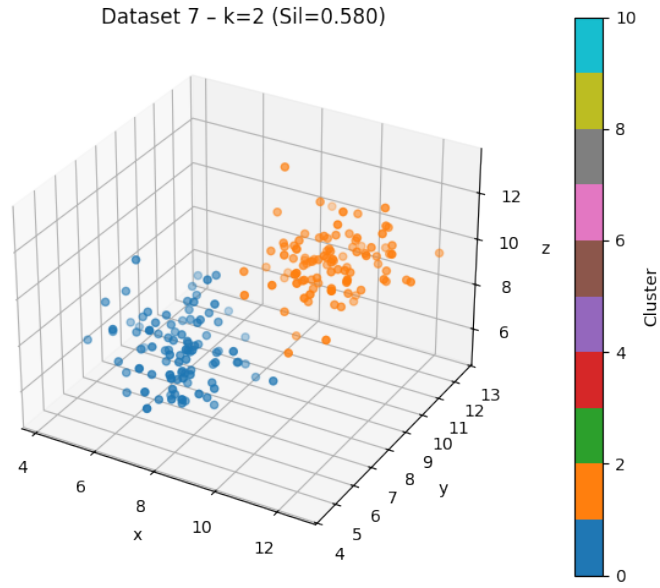


Figure 23: Dataset 7 Clustering with k Clusters

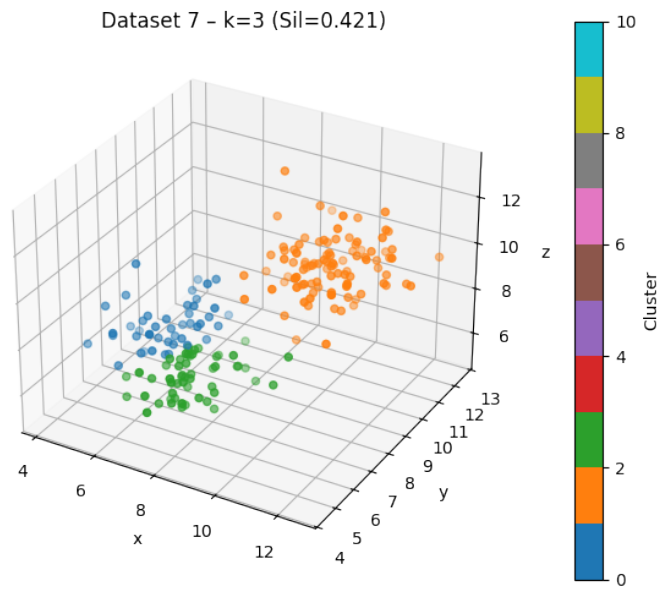


Figure 24: Dataset 7 Clustering with $k + 1$ Clusters